

Entity Decomposition with Filtering: A Zero-Shot Clinical Named Entity Recognition Framework

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Abstract

Clinical named entity recognition (NER) aims to retrieve important entities within clinical narratives. Recent works have demonstrated that large language models (LLMs) can achieve strong performance in this task. While previous works focus on proprietary LLMs, we investigate how open NER LLMs, trained specifically for entity recognition, perform in clinical NER. Our initial experiment reveals significant contrast in performance for some clinical entities and how a simple exploitation on entity types can alleviate this issue. In this paper, we introduce a novel framework, entity decomposition with filtering, or EDF. Our key idea is to decompose the entity recognition task into several retrievals of entity sub-types and then filter them. Our experimental results demonstrate the efficacies of our framework and the improvements across all metrics, models, datasets, and entity types. Our analysis also reveals substantial improvement in recognizing previously missed entities using entity decomposition. We further provide a comprehensive evaluation of our framework and an in-depth error analysis to pave future works.

1 Introduction

Clinical narratives hold immense value for clinical experts (Tayefi et al., 2021; Mahbub et al., 2022; Raghavan et al., 2014; Rannikmäe et al., 2021), largely due to their wealth of information often inaccessible in the structured data of the electronic health records (EHR) (Mahbub et al., 2023; Goodman-Meza et al., 2022; Kharrazi et al., 2018; Rannikmäe et al., 2020; Hernandez-Boussard et al., 2019; Boag et al., 2018). Their free format, however, causes significant challenges for healthcare systems to utilize. The richness of information trapped within the narratives, followed by its significance, has spurred a plethora of works in tackling the *clinical information extraction* problem within

the clinical NLP community (Wang et al., 2018; Landolsi et al., 2023).

One key building block in clinical information extraction is named entity recognition (NER), focusing on identifying clinical concepts within these narratives. Prior methods (Wang et al., 2018) rely on either traditional natural language processing (NLP) techniques or supervised learning methods. Nevertheless, the former approach can be fragile, while the latter requires significant effort to annotate. In addition, supervised methods cannot simply scale for the large number of concepts available in the clinical domain (Bodenreider, 2004).

In light of this, Large Language Models (LLMs), with their strong capabilities for zero- and few-shot learning (Chowdhery et al., 2023; Brown et al., 2020; Thoppilan et al., 2022; Touvron et al., 2023), serve as promising solutions for clinical NER. While previous works focus on LLMs trained on general tasks (Agrawal et al., 2022; Liu et al., 2023; Gero et al., 2023), here we focus on LLMs specifically trained for entity recognition, or *open NER LLMs* (Zhou et al., 2023; Ding et al., 2024). Inspired by their results in clinical domain (Zhou et al., 2023), outperforming even proprietary LLMs (Brown et al., 2020), we conduct deeper investigations in this study. Surprisingly, our preliminary experiment (Section 5.1) suggests a stark performance gap between retrieving different clinical entities (Figure 2). For instance, an open NER LLM called UniversalNER (Zhou et al., 2023) performs significantly better at extracting medications rather than clinical treatments (85.88% vs 53.81% Exact Match F1-scores). Upon closer inspection, we find these unidentified treatment entities can be effectively recognized by exploiting *simpler, albeit specific*, entity types. For example, by explicitly specifying “medication” rather than “treatment” as input, the model can capture a substantial portion of the previously unidentified medication-related treatment entities.

Building upon this insight, we present a novel framework, *entity decomposition with filtering*, or EDF, aimed at tackling clinical NER. To the best of our knowledge, we are the first to explore strategies to effectively use open NER LLMs in the clinical domain *without using any samples*. We draw inspiration from the divide-and-conquer paradigm (Knuth, 1998), which breaks down a complex problem into simpler sub-problems. Concretely, we posit that a direct retrieval of entities may be too complex for the model and instead propose to disentangle it into a series of retrievals through *entity decomposition*. Unlike the previous approach (Xie et al., 2023), entity decomposition breaks down the task by identifying through their *entity sub-types*, which, ideally, are easier to retrieve. Nonetheless, entity decomposition alone is insufficient since some entity sub-types do not form strict subsets (further discussed in Section 3.2.3). To address this, we introduce a *filtering* mechanism in our framework to further improve performance. We illustrate them in Figure 1.

Our work introduces a cost-effective approach to improve clinical entity recognition using open-source large language models. Overall, we observe improvements across models, metrics, datasets and entity types. Our ablation study also reveals the robustness of our framework, allowing users to adjust the components based on their performance and cost.

2 Related Work

2.1 LLMs for Clinical NER

LLMs are promising for many clinical tasks (Singhal et al., 2023; Agrawal et al., 2022; Clusmann et al., 2023). Concurrently, several works aim to improve their performance on clinical NER. For instance, Agrawal *et.al.* (Agrawal et al., 2022) proposes a guided prompt design along with a resolver to handle the structured output space required by NER, while others (Hu et al., 2024, 2023; Liu et al., 2023) use prompt engineering. Outside the clinical domain, several works tackle NER either by framing it as a sequence labeling task (Wang et al., 2023), using label decomposition and syntactic augmentation (Xie et al., 2023), or improving the structured label space (Li et al., 2024), similar to Agrawal *et.al.* (Agrawal et al., 2022). Most of these works focus on LLMs trained in handling diverse tasks such as ChatGPT (Brown et al., 2020). In contrast, we focus on open NER LLMs (Zhou

et al., 2023; Ding et al., 2024), which have two key differences. First, they are trained specifically for entity recognition tasks and do not require structured output space handling (Agrawal et al., 2022; Li et al., 2024). Second, their instruction-tuning mostly focused on the diversity of entities rather than the instructions (e.g., keeping the prompt constant), which may limit the efficacy of prompt engineering techniques. Furthermore, unlike previous works (Hu et al., 2024, 2023; Liu et al., 2023), *our work does not fall under prompt engineering*. Notably, prompt engineering is limited to prompt-based models, while our work is model-agnostic and, thus, is applicable to BERT-based models (Zaratiana et al., 2024).

2.2 Task Decomposition in LLMs

The idea of task decomposition, solving complex tasks through solving its constituent simpler sub-tasks, can be dated back to (Lazarou et al., 1998). Previous works propose task decompositions for LLMs to tackle complex problems (Zhou et al., 2022; Xie et al., 2023). Concurrent with our work, Xie *et.al.* (Xie et al., 2023) suggests decomposing NER into a multi-turn dialogue, asking the model one question for each label. However, some open NER LLMs (Zhou et al., 2023) can only extract one label at a time, thus limiting the efficacy of Xie *et.al.* (Xie et al., 2023). Here, *we propose to decompose NER on entity-level rather than label-level*. Concretely, we can further decompose each label into simpler labels. Our method also complements Xie *et.al.* (Xie et al., 2023) since these decompositions can be performed sequentially. Besides, our work aims to improve open NER LLMs, which have several key differences from other LLMs as briefly discussed in Section 2.1

2.3 Open NER LLMs

Clinical narratives fall under domains with a large number of concepts and scarce annotations. Thus, developing open named entity recognition LLMs (Zhou et al., 2023; Ding et al., 2024) is timely and crucial research for clinical NER. Despite the progress, existing works focus on training the backbone models. Furthermore, these models present a unique challenge and cannot be treated similarly to other LLMs (Section 2.1). Our work paves a way to adapt them for clinical domains without finetuning.

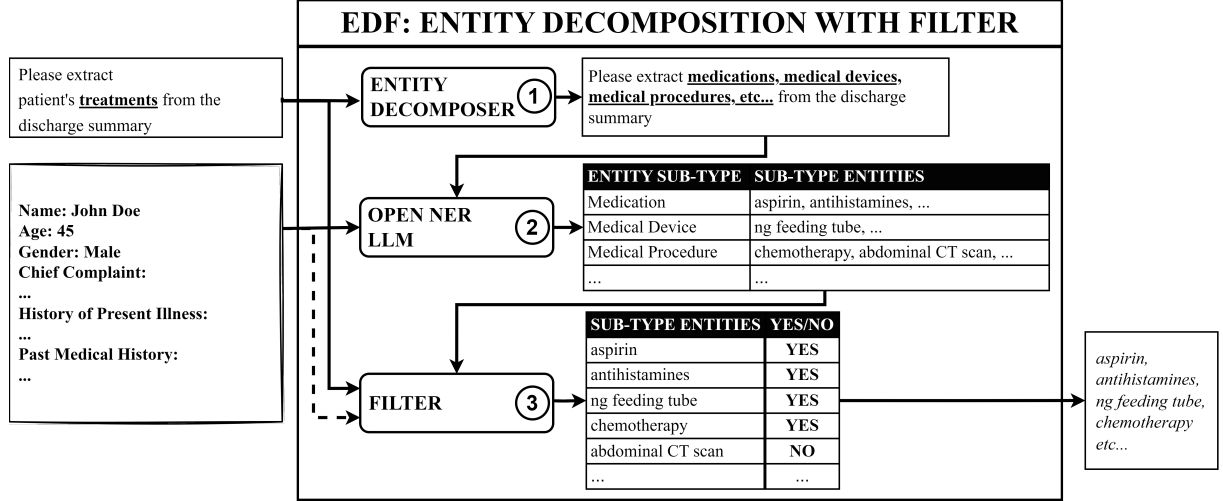


Figure 1: **Entity Decomposition with Filtering.** Our novel framework is composed of three components: (1) **Entity Decomposer** breaks down the target entity type (e.g., treatment) into several entity sub-types (e.g., medication, medical device, medical procedure, etc), (2) **Open NER LLM** generates the sub-type entities, (3) **Filter** removes sub-type entities outside the target entity type. See Section 3 for details.

3 Method

3.1 Problem Definition

Clinical narrative holds important entities about a patient’s medical history. In this work, we aim to tackle clinical NER, focusing on extracting them. We frame the problem through the lens of a text-generative task. Let $\mathbf{x}$ be a clinical narrative, and further let $\mathbf{t}$ be the *target entity type* we want to extract. We aim to retrieve the *target entities set* $\mathcal{Y}$ corresponding to $\mathbf{t}$ from $\mathbf{x}$. To illustrate, if $\mathbf{x}$ is a patient’s discharge summary and $\mathbf{t} = \text{“medication”}$, then the goal is to extract medication entities in the discharge summary. In this case, the output can be $\mathcal{Y} = \{\text{“aspirin”}, \text{“methanol”}, \dots\}$.

3.2 Entity Decomposition with Filtering

As introduced in Section 1, directly retrieving the target clinical entities may be too challenging, particularly for models without domain-specific training, such as open NER LLMs. We propose to break the task into multiple retrievals of *sub-type entities* instead. We define sub-type entities as entities belonging to a sub-type of the target entity type. Concretely, let $\hat{\mathcal{Y}}_i$ be a set of sub-type entities corresponding to the i -th *entity sub-type* s_i and let $\hat{\mathcal{Y}}$ be our predicted complete set, where ideally $\hat{\mathcal{Y}}_i \subseteq \hat{\mathcal{Y}}$. The first part of our framework, *entity decomposer*, aims to iteratively collect $\hat{\mathcal{Y}}_i$ to produce $\hat{\mathcal{Y}}$, $\bigcup_{i=1}^N \hat{\mathcal{Y}}_i = \hat{\mathcal{Y}}$. The last part of our framework, *filter*, involves removing sub-type entities in $\hat{\mathcal{Y}}$ outside the target entity type $\mathbf{t}$. The filtered version $\hat{\mathcal{Y}}_f$ then serves as the final output. That is, $\mathcal{Y} = \hat{\mathcal{Y}}_f$. We provide more details of our framework in the

following sections. Figure 1 presents the overall architecture of our framework, entity decomposition with filtering, or EDF.

3.2.1 Entity Decomposer

The first step in our framework is to identify what constitutes entity sub-types for the target entity type $\mathbf{t}$. Let $\mathcal{D}(\mathbf{t}) = \mathcal{S}$ be the *entity decomposer* module, aimed to produce a set of N *entity sub-types* $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ using the target entity type $\mathbf{t}$. To illustrate how the module works, let’s take “treatment” as an example. In a patient’s discharge summary, “treatment” constitutes a myriad of entities, including medications, medical procedures, and more (see Figure 1). In this case, $\mathcal{D}$ aims to decompose $\mathbf{t} = \text{“treatment”}$ into $\mathcal{S} = \{\text{“medication”}, \text{“medical procedure”}, \dots\}$.

There are several ways to construct this module, and clinical practitioners can resort to different methods based on costs and performances. Some examples include manually curating the entity sub-types (possibly involving clinical experts) or obtained using existing tools such as a medical knowledge base (Bodenreider, 2004). We provide details in Appendix A

3.2.2 Open NER LLM

After defining the entity sub-types $\mathcal{S}$, the next step is to retrieve the sub-type entities $\hat{\mathcal{Y}}$. In this work, we leverage an *open NER LLM*. We base its formal definition on previous works (Zhou et al., 2023). Let R be an open NER LLM tasked to retrieve the sub-type entities $\hat{\mathcal{Y}}_i$ of s_i from the clinical narrative $\mathbf{x}$. That is, $R(\mathbf{x}, s_i) = \hat{\mathcal{Y}}_i$. To construct the complete set $\hat{\mathcal{Y}}$, the model would iteratively collect

$\hat{\mathcal{Y}}_i$ from each s_i . This may be of concern, given that it requires multiple iterations. To address this, we also consider other variants (Ding et al., 2024; Zaratiana et al., 2024) of open NER LLMs, which we formally define as $R^*(\mathbf{x}, S) = \hat{\mathcal{Y}}$. In contrast to R , these variants are capable of simultaneously extracting multiple entity types at once, thereby reducing the whole iterative sub-type entity retrievals to only one forward pass.

We base the use of an open NER LLM on several key reasons. First, open NER LLMs are versatile in recognizing arbitrary entities, which is a characteristic of sub-type entities (e.g. they are not bounded to a predefined label set). Second, in contrast to other LLMs, open NER LLMs are explicitly trained for NER tasks, substantially reducing the effort of mapping LLM generative output to the structured output space of NER (Agrawal et al., 2022). In addition, our preliminary experiment in Section 5.1 suggests that open NER LLMs perform surprisingly well in recognizing basic clinical entities such as medications, making them strong candidates for this module.

3.2.3 Filter

We discuss the last step of our framework here. Formally, let $f(\hat{\mathcal{Y}}, t, C) = \hat{\mathcal{Y}}_f$ be the *filter* module, where C denotes a *context* and $\hat{\mathcal{Y}}_f \subseteq \hat{\mathcal{Y}}$. In this framework, f can be viewed as a binary classifier, assigning a positive label if the entity corresponds to t , and a negative label otherwise. The filter module then aggregates and outputs the positively classified entities. Overall, f aims to eliminate sub-type entities within $\hat{\mathcal{Y}}$ that do not fall under the target entity type t based on its context C (e.g. the paragraph the entity exists). Take “treatment” and “medical procedure” for example. While “treatment” includes “medical procedure”, not all “medical procedure” qualify as a “treatment”. For instance, some medical procedures (e.g., endoscopy) may serve purely for diagnosis purposes and should not be classified as “treatment.”

We provide the rationale behind using a context C in the filter module. Unlike general domain entities, clinical entities can largely rely on context cues. To illustrate, consider “adverse drug event (ADE)” (Zed et al., 2008; Lazarou et al., 1998). By definition, “ADE” is an “injury resulting from a medical intervention” (Henry et al., 2020). Thus, one of its sub-types may be “injury”. However, for a filter to dictate whether an injury corresponds to “ADE”, it needs to be aware of where the injury

comes from. Injuries due to an accidental fall may not be an ADE if the patient does not have any medical interventions. In other words, *the filter requires the context in which the entity occurs to provide an accurate prediction*. Moreover, our experiment suggests that while some clinical entities do not need contextual information, they can benefit from it (see Section 5.3.3).

4 Experimental Setup

We provide the experimental setup here and leave the details in the Appendix. All of the base models are available in huggingface*.

4.1 Open NER LLMs

We take SOTA open NER LLMs in our experiments and further improve them. Per definition in Section 3.2.2, they may be categorized based on how many entity types can be extracted simultaneously. To this end, we use **UniversalNER** (Zhou et al., 2023) and **GNER** (Ding et al., 2024) as the representative for R and R^* , respectively. Concretely, we use UNIVERSALNER-TYPE-7B and GNER-LLAMA-7B. Both are finetuned on the PileNER dataset (Zhou et al., 2023), which is generated from GPT 3.5 (Brown et al., 2020). We only experiment with their default prompts since, in contrast to other LLMs, open NER LLMs are trained on a set of diverse entity types rather than prompts (Zhou et al., 2023)

4.2 Entity Decomposers

Here, we experiment with different techniques to decompose entities. **First**, given that clinical narratives require specialized knowledge, we consider entity sub-types curated by clinical experts. Specifically, we take the annotation guidelines available from the datasets. **Second**, we use ChatGPT (Brown et al., 2020) to decompose clinical entities automatically. We draw our inspiration from the recent success of ChatGPT in the clinical domain (Agrawal et al., 2022; Singhal et al., 2023). Furthermore, using ChatGPT for entity decomposition is more cost-effective and scalable. **Third**, we utilize the Unified Medical Language System (UMLS) (Bodenreider, 2004), a medical knowledge base, for retrieving entity sub-types. We provide more details in the Appendix A.

*<https://huggingface.co>

4.3 Filters

We use **Asclepius** (Kweon et al., 2023) and **LlaMA-2** (Touvron et al., 2023) trained on clinical and general domains, respectively. Specifically, we use ASCLEPIUS-LLAMA2-7B and LLAMA-2-CHAT-7B versions. Given the inherent generative nature of LLMs, we restrict their outputs to "Yes/No" responses (when applicable) using grammar-constrained decoding (Geng et al., 2023). This strategic constraint reduces the number of generated tokens, resulting in increased inference speed. By default, we prompt the model by asking "Can {entity} be considered as a/an {entity_type}?". We try different prompts in Section 5.3.4. For entities that require context, we use a simple preprocessing method so that the context provides sufficient information to extract the clinical entities. That is, we include the whole paragraph (or more) in which the entity occurs from the clinical narrative.

4.4 Datasets

We focus our experiments on extracting concepts from publicly available clinical notes[†]. We use **ClinicalIE** (Agrawal et al., 2022), **i2b2 2010** (Uzuner et al., 2011), **i2b2 2012** (Sun et al., 2013), **i2b2 2018 Task 2** (Henry et al., 2020) and **CLEF 2014** (Mowery et al., 2014) datasets in this paper. They are available in Harvard DBMI[‡] for i2b2 datasets, PhysioNet[§] (Goldberger et al., 2000) for CLEF 2014 and huggingface for ClinicalIE. In these datasets, context is not required to identify clinical entities except for the i2b2 2018 dataset. We provide further details in the Appendix B.

4.5 Baselines and Metrics

Given the scarce methods to compare with, we use the method developed in Xie et al. (Xie et al., 2023) as our baseline. Concretely, we extract each entity type one at a time using UniversalNER and GNER. We also compare with UNIVERSALNER-ALL, trained with both PileNER and over 40 supervised datasets, including our experiment sets. We use *Precision* (P), *Recall* (R), and *Exact Match F1-Score* (F1) as evaluation metrics, similar to previous works.

[†]All datasets are available from the providers under appropriate data usage agreements

[‡]<https://portal.dbmi.hms.harvard.edu/>

[§]<https://physionet.org/>

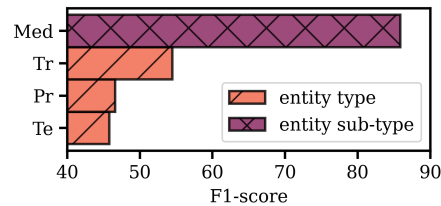


Figure 2: Open NER LLM (UniversalNER) performs better at extracting entity sub-type (Med) rather than the target entity type (Tr, Pr, Te). We use ClinicalIE for Med (medication) and i2b2 2012 for Tr (treatment), Pr (problem), and Te (test). Discussion in Section 5.1

5 Experimental Results

Throughout this section, we abbreviate the entity types in the tables as follows to save space: Tr for treatment, Pr for problem, Te for test, CD for clinical department, DD for disease/disorder, AD for adverse drug, and ADE for adverse drug event.

Given the space limit, we include more results in the Appendix, including few-shot (Appendix E), performance on a BERT-based model (Appendix F), and error analysis on CD performance drop (Appendix G).

5.1 Preliminary Experiment

We conduct a preliminary experiment to confirm that open NER LLMs perform better at recognizing sub-type entities rather than the target entity types. For target entities, we use the i2b2 2012 dataset, which contains decomposable entity types (i.e., entities can further be divided into sub-type entities). For sub-type entities, we use ClinicalIE, a medication extraction dataset.

Figure 2 illustrates the result and confirms our hypothesis. That is, it is harder to recognize the target entity types (that are decomposable) compared to the sub-type entities. For instance, we observe a stark difference between “medication” extraction and “treatment” extraction, where the former is an entity sub-type of treatment.

5.2 Overall Performance

We present our results in Table 1 and Figure 3. For detailed numbers on precision and recalls, we leave them in the Appendix C.

On average, EDF outperforms baseline by 2.54% and 5.82% F1-score on UniversalNER and GNER, respectively. Interestingly, for some entity types (e.g. treatments and tests in i2b2 2010), GNER performs similarly or even outperforms UniversalNER. This suggests that models that can recognize multiple entities simultaneously can benefit more from using our framework.

Table 1: F1-score performance (%) comparison between baseline (B) from Xie *et.al.* (Xie et al., 2023), Entity Decomposer (ED) only, Filter (F) only, supervised (UniNER-all) and EDF (Ours) methods across datasets, entity types, and models. We use Asclepius and the default prompt strategy (Section 3.2.3) for the filter and annotation guideline for the entity decomposer. We mark the best results in **bold** and second-best in underlined. Discussion in Section 5.2

Dataset	Entity Type	UniNER					GNER					UniNER-all (Supervised)
		B	ED	F	EDF	Δ	B	ED	F	EDF	Δ	
i2b2 2010	Tr	53.81	43.56	<u>54.35</u>	59.63	+5.82	53.31	39.43	<u>62.72</u>	63.23	+9.92	74.95
	Pr	49.71	41.99	52.77	<u>51.46</u>	+1.75	40.40	35.91	50.77	<u>50.62</u>	+10.22	73.11
	Te	48.78	36.99	40.58	<u>44.17</u>	-4.61	<u>37.27</u>	36.40	37.23	45.78	+8.51	72.43
i2b2 2012	Tr	<u>54.49</u>	48.25	52.82	58.25	+3.76	50.38	40.16	<u>55.63</u>	57.09	+6.71	72.37
	Pr	46.61	42.45	<u>49.64</u>	50.24	+3.63	41.11	38.00	<u>47.15</u>	48.82	+7.71	75.16
	Te	45.78	31.52	43.98	46.34	+0.56	33.17	27.55	<u>41.09</u>	46.66	+13.49	65.47
	CD	41.44	32.57	34.51	<u>38.66</u>	-2.78	58.88	20.46	<u>39.20</u>	37.74	-21.14	44.37
CLEF 2014	DD	<u>46.73</u>	45.94	45.26	58.25	+11.52	18.06	<u>21.11</u>	18.44	26.91	+8.85	63.19
i2b2 2018	AD ^{†‡}	19.07	8.59	27.36	<u>24.92</u>	+5.85	3.16	4.54	10.08	13.66	+10.50	14.37
	ADE [†]	<u>9.56</u>	2.64	15.90	9.43	-0.13	0.61	1.09	<u>2.40</u>	4.05	+3.44	31.18
Avg.		41.60	33.45	<u>41.72</u>	44.14	+2.54	33.64	26.47	<u>36.47</u>	39.46	+5.82	58.66

[†] AD and ADE are entity types that require context [‡] UniNER-all is not trained to extract AD entities

Entity decomposition (ED) improves recall but decreases precision. As illustrated in Figure 3, we observe a consistent improvement in recall across diverse datasets and entity types for both models, suggesting that entity decomposition facilitates the identification of previously missing entities while being robust to the backbone models. For precision, however, we notice a drop in performance. As discussed in Section 3.2.3, some sub-type entities may not form a subset of the target entities, causing performance degradation on precision. Further examination of the F1-score reveals a decline in overall performance, which justifies the necessity of incorporating a filtering mechanism.

Conversely, filtering (F) benefits precision but degrades recall. This shows contrasting results compared to entity decomposition across datasets, entity types, and models. The overall F1-score improvement on EDF compared to using each component individually suggests that entity decomposition and filtering complement each other. This emphasizes the necessity to incorporate both of them.

EDF is robust to out-of-distribution entities compared to supervised training. We want to emphasize that *our method does not require any training* as opposed to the supervised approach. We use UniversalNER+EDF and UniversalNER-all as comparison. Despite the performance gap, we observe that on entities not covered in the training label set (e.g., adverse drug or AD), EDF outperforms by more than 10% on the F1-score. This shows the robustness of our method.

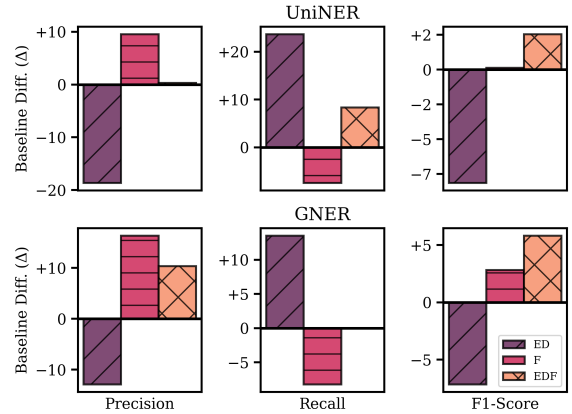


Figure 3: Average performance improvement to baseline across metrics and models. Entity decomposition (ED) improves recall but degrades precision. Filter (F) increases precision but decreases recall. Our method (EDF) achieves better performance overall. Discussion in Section 5.2

5.3 Ablation Study

We perform additional experiments to test the efficacy of our framework using different entity decomposers or filters. We focus on the overall performance or F1-score. To reduce the cost of our experiments, we only experiment with the i2b2 2012 dataset, given their diversity in entity types. Unless otherwise specified, we use UniNER as the open NER LLM, annotation guideline for the entity decomposer module, and Asclepius for the filter.

5.3.1 Entity Decomposers

First, we conduct ablation study with different entity decomposers as described in Section 4.2 and present our results in Table 2. We do not experiment with ChatGPT and UMLS for the “clinical department” entity type since (1) ChatGPT is un-

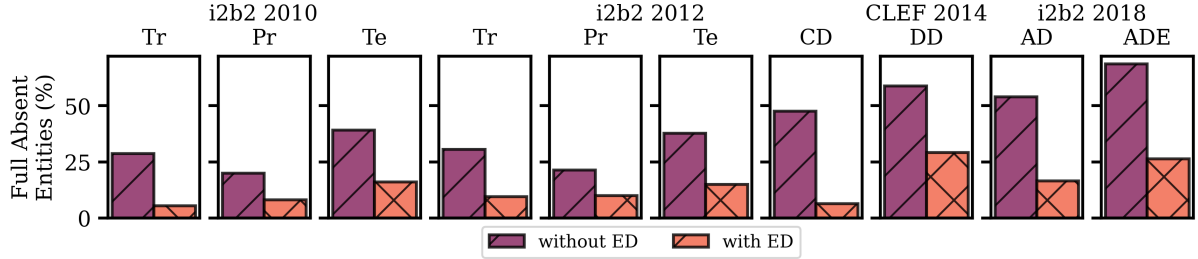


Figure 4: Entity Decomposition (ED) captures most previously fully absent entities. Lower value is better. Discussion in Section 5.4.1

Table 2: Ablation Study on Entity Decomposer.

Entity Decomposer	Filter	Entity Type			
		Tr	Pr	Te	CD
Baseline		54.49	46.61	45.78	41.44
Annotation	✗	48.25	42.45	31.52	32.57
ChatGPT		48.62	44.05	43.80	-
UMLS		55.57	46.98	44.68	-
Annotation	✓	58.25	50.24	46.34	38.66
ChatGPT		58.09	51.12	44.83	-
UMLS		58.23	51.04	44.71	-

able to produce reasonable entity sub-types and (2) we find there are no correspondence semantic types in UMLS.

EDF is robust to entity decomposer module.

We observe competitive performance between a general LLM, medical knowledge base and clinical experts. For instance, on treatment entities, the performance between clinical experts and ChatGPT is 58.25% vs 58.09%. Thus, even without curation from clinical experts, our method can achieve competitive results.

Without filters, UMLS outperforms other entity decomposers. Most entity sub-types in UMLS form subsets to the target entities; hence, a filter may not be necessary in contrast to other entity decomposers. Interestingly, for some entities, it performs better than vanilla prompt (e.g. treatments and problems). This is significant since even without filtering, our framework, specifically entity decomposition, can still outperform the vanilla approach with proper curation of entity sub-types. We remark that this does not take away the value of a filter module. For instance, UMLS can benefit more from integrating it, as shown in Table 2.

5.3.2 Filter Models

Here, we investigate how different filter models affect the overall performance of our framework. Specifically, we compare domain-specific and general LLMs. We present the results in Table 3

Clinical model is better at recognizing entities requiring clinical expertise. Specifically, we ob-

Table 3: Ablation Study on Filter Model.

Entity Decomposer	Filter Model	Entity Type			
		Tr	Pr	Te	CD
Baseline		54.49	46.61	45.78	41.44
✗	Asclepius	52.82	49.64	43.98	34.51
	Llama2	51.90	47.49	45.27	35.37
✓	Asclepius	58.25	50.24	46.34	38.66
	Llama2	53.94	45.55	37.41	38.69

Table 4: Ablation Study on Filter Context.

Entity Decomposer	Filter Context	Entity Type			
		Tr	Pr	Te	CD
Baseline		54.49	46.61	45.78	41.44
✗	none	52.82	49.64	43.98	34.51
	sentence	53.85	42.90	43.91	37.18
	document	54.67	41.53	44.45	44.91
✓	none	58.25	50.24	46.34	38.66
	sentence	60.58	45.17	46.68	40.80
	document	58.24	42.24	45.31	48.90

serve that it is superior to a general domain model in “treatment”, “problem”, and “test” entities. For “clinical department”, however, they perform similarly. This is unsurprising since the former often requires clinical-specific knowledge compared to the latter.

5.3.3 Filter Context

As discussed in Section 3.2.3, some clinical entities need context. Here, we investigate whether context helps for entities not requiring it. We compare the performance between filtering (1) without context, (2) including the sentence the entity appears in, and (3) providing the whole clinical narrative or document. We show the results in Table 4.

Context may improve or hurt performance.

Overall, we observe mixed results across different entity types, with and without an entity decomposer. For instance, we observe slight improvements for “treatment” and “test” entities. On the other hand, the performance of the “problem” entity consistently drops the more context we provide. We provide further analysis in Section 5.4.2.

Table 5: Ablation Study on Filter Prompt

Entity Description	Entity Type			
	Tr	Pr	Te	CD
$\times$	58.25	50.24	46.34	38.66
$\checkmark$	50.32	47.38	33.18	45.97

5.3.4 Filter Prompts

LLMs are often brittle to prompting strategies (Zhu et al., 2023). We experiment with how incorporating the entity description into the filter affects our framework’s performance. For “treatment”, “problem” and “test”, we use descriptions available in i2b2 2010 annotation guidelines. For “clinical department”, we provide our own definition. We put the prompts in Appendix D and present the results in Table 5.

Complex entity description degrades performance. Our experiment shows a notable performance drop in all entity types except “clinical department”. We hypothesize that the descriptions provided in the guidelines may be too complex for the model to understand. In contrast, we observe more than 7% F1-score improvement on “clinical department”, which uses our handcrafted and concise definition.

5.4 Error Analysis

5.4.1 Entity Decomposition Missing Entities

Despite the significant improvement in recall through entity decomposition, some entities remain unrecognizable. Thus, we analyze the potential sources of these errors. First, we calculate the percentage of entities *fully absent* from the predictions. To illustrate, if the label is “his aspirin” and the prediction is “aspirin”, we do not deem it fully absent since the prediction partially captures the label. Figure 4 illustrates the percentage of fully absent entities for each entity type in the dataset.

Entity decomposition significantly reduces the number of fully absent entities. For instance, only 5.5% entities are fully absent for “treatment” in the i2b2 2010 dataset after entity decomposition. We observe improvements across all entity types.

The majority of fully absent entities are abbreviations and homonyms. For example, open NER LLM cannot capture “CVA”, an abbreviation for “cerebral vascular accident”, after entity decomposition. Another example is “delivery”, which carries nuanced meanings in different contexts (e.g. childbirth or route of medications). Furthermore, certain entities such as “HD” are both abbreviations and homonyms (i.e., high definition vs hemodialysis).

Table 6: i2b2 2012 Polarity Dataset Statistics

Polarity	Entity Type (# of samples)			
	Tr	Pr	Te	CD
Positive	3684	4164	2544	996
Negative	145	858	52	0

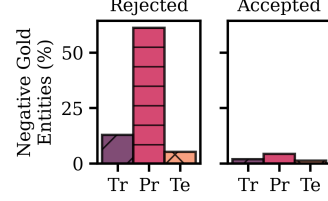


Figure 5: The majority of the rejected golden “Problem” entities are negative, leading to a performance drop in Table 4 when using context. Further discussion in Section 5.4.2.

5.4.2 Performance Drop using Context

Section 5.3.3 shows that there is a notable performance degradation for “problem” when context is provided. To investigate, we observe the ground-truth “problem” entities that are removed by the filter. Interestingly, we find that for most of them, the context specifically stated that the patient does not experience these problems. We then conduct further investigation on their *polarity* attribute, which contains information on whether the patient is experiencing medical problems (or taking certain medications, for instance). To clarify, if there are explicit mentions that a patient does not have certain medical problems, the polarity is negative. Otherwise, it is positive. We conduct an analysis of how entity polarity affects filter responses. We plot our results in Figure 5.

The “negative” polarity degrades performance. First, the dataset statistics in Table 6 show that almost a fifth of “problem” entities are “negative”, making it likely that these gold labels would be rejected. Furthermore, Figure 5 reveals that the “negative” polarity causes the performance drop on the “problem” entity, as revealed by how a majority of the rejected gold “problem” entities are “negative”.

6 Conclusion and Future Work

In this work, we propose a novel EDF framework to tackle the clinical named entity recognition task. Our comprehensive experiments demonstrate the strength of our framework across different dimensions. We also thoroughly investigate each framework component and provide several key insights. In future works, we hope to explore how to address the limitations of our work described in Section 7.

7 Limitations

First, we restrict our work to clinical narratives and have yet to explore how our framework generalizes to other texts. In this work, we deliberately focus on how well the method generalizes to different datasets, which (1) tackle different and clinically significant (Lehman et al., 2022) entity types, (2) are collected from different institutions (thus different distributions), (3) are de-identified in different ways (e.g., masks used for the patient’s sensitive information), (4) used different formats (e.g., header names and section organizations), etc. In fact, each patient is a unique case, and each of them can be treated as a separate domain (Yang et al., 2023). Thus, generalizing to these datasets is already a significant challenge. However, testing how well our framework generalizes beyond clinical narratives would be an interesting avenue. Note that our motivation for this framework is that we found some clinical entities are easier to identify through simpler terms. This is particularly true for clinical narratives since most entities that are of interest (Lehman et al., 2022) follow this assumption. Thus, we designed our framework based on the characteristics of entities inside clinical narratives, not the narratives themselves. This is the reason we hypothesize that our framework may work outside clinical narratives (with similar entity characteristics). We leave this to future work.

Second, we restrict our work to only open-sourced models and leave experiments on proprietary models to future works. Most publicly available clinical narratives are under restrictive licenses. Hence, we cannot simply use commercial models, which may leak the data to a third party that is not covered by the restricted data use agreements. In contrast, open-sourced models have more practical values (e.g., they can be deployed). In this work, we deliberately use strong open-sourced models such as UniversalNER (Zhou et al., 2023), which performs better than ChatGPT (Brown et al., 2020). However, how open-sourced models fare with other proprietary models on clinical NER is still unknown. We leave them to future works.

Third, our work falls under the healthcare domain, a high-stakes setting. Despite the good performance of our methods, there is still a long way to reach the performance achieved in the original challenge of the datasets years ago, which are from the supervised learning settings. We provide their results in [Appendix K](#). It is also critical that NER

performance in clinical texts can satisfy the high requirements set by healthcare applications. Nevertheless, our work paves a potential solution for the zero-shot clinical named entity recognition task.

Fourth, despite our attempts in generalization across datasets, it is very unfortunate that existing publicly available annotated clinical datasets only contain a few entity types. The issue is exacerbated by the significance of clinical domain, with emphasize on patient data protection (thus harder to publicly share) and the domain expertise to annotate. Once datasets with other entity types are made available, we will apply our methods to other entity types, which we are not able to do at this point given the lack of datasets with other entity types.

Fifth, our framework may provide higher inference cost than supervised methods. We remark that we develop a framework for a high-stakes domain that imposes high-performance requirements. Furthermore, our method is much more cost-efficient than curating an annotated datasets, which is required for supervised methods. Our framework is also flexible such that it provides trade-off between performance and inference cost depending on the user’s needs.

8 Ethic Statement

Our research is conducted on open, retrospective clinical datasets without human subject intervention and thus will not harm human subjects. Furthermore, the clinical domain is complex and requires evaluation beyond performance, particularly regarding safety and bias. Unfortunately, the clinical narratives in our datasets are not associated with specific patients, impeding such evaluations. Further evaluations and validations from clinical experts will be needed to translate research into the clinical decision process.

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Appendix

A Entity Decomposers

We provide the details for each of our entity decomposition methods described in [Section 4.2](#) here:

- **Manually curating a set of candidate types using expert-level knowledge.** Here, we refer to the annotation guidelines available in existing datasets since we believe they are curated by domain experts. For “Tr”, “Pr”, “Te” and “DD” we take the annotation guidelines from i2b2 2010. For “CD”, we use i2b2 2012. For “AD” and “ADE”, we use i2b2 2018 Task 2. We list the curated set in [Section A.1](#).
- **Prompting an LLM for automatic generation.** We prompt ChatGPT with “*You are an intelligent clinical language model. Your job is to extract {entity_type} from a patient’s discharge summary. What entities can be considered as {entity_type} in a discharge summary?*” for each entity type. For reproducibility, we present the results in [Section A.2](#).
- **Utilizing an existing medical knowledge bank.** We use the Unified Medical Language System (UMLS) since it contains standardized medical vocabulary for many clinical entities. Here, we take the UMLS semantic types for “Tr”, “Pr” and “Te” available in i2b2 2010 guidelines. We list the curated set in [Section A.3](#).

A.1 Annotation

Treatment: medical treatment, medical intervention, medical procedure, medical device, treatment, biological substance, drug, medication

Problem: medical problem, disease, syndrome, symptom, medical condition, behavior, virus, bacterium, injury, abnormality, abnormal test result, mental status

Test: medical test, medical procedure, medical panel, medical examination, medical evaluation, test, procedure, laboratory procedure, diagnostic procedure, panel, measure, physiologic measure, vital sign, examination, evaluation

Clinical Department: clinical department, medical department, clinical unit, clinical service, clinical

Table 7: Dataset Statistics

Dataset	# samples
i2b2 2010	27,625
i2b2 2012	7,446
i2b2 2018	9,181
CLEF 2014	10,422

practice, clinical room, department, location, building, hospital

Disease/Disorder: medical problem, disease, syndrome, symptom, medical condition, behavior, virus, bacterium, injury, abnormality, abnormal test result

Adverse Drug: drug

Adverse Drug Event: medical problem

A.2 ChatGPT

Treatment: medical treatment, medication, medical procedure, therapy, medical intervention, consultation, counseling, discharge instruction, supportive care

Problem: medical problem, medical diagnosis, disease, abnormal test result, symptom, abnormal imaging finding, complication, chronic health condition, medication side effect, mental health issue, social determinants of health

Test: medical test, laboratory test, imaging study, diagnostic procedure, genetic test, electrodiagnostic test, functional test, microbiological test

A.3 UMLS

Treatment: medical treatment, therapeutic procedure, preventive procedure, medical device, steroid, pharmacologic substance, biomedical material, dental material, antibiotic, clinical drug, drug delivery device

Problem: medical problem, pathologic function, disease, syndrome, mental dysfunction, behavioral dysfunction, cell dysfunction, molecular dysfunction, congenital abnormality, acquired abnormality, injury, poisoning, anatomic abnormality, neoplastic process, virus, bacterium, symptom

Table 8: Extension of Table 1 for Precision (%).

Dataset	Entity Type	UniNER					GNER					UniNER-all (Supervised)
		B	ED	F	EDF	Δ	B	ED	F	EDF	Δ	
i2b2 2010	Tr	51.63	30.35	61.71	55.09	+3.46	46.08	27.19	74.16	66.60	+20.52	80.63
	Pr	44.95	30.94	56.02	47.15	+2.20	33.71	25.61	55.43	47.63	+13.92	75.87
	Te	53.51	26.04	58.67	48.84	-4.67	32.70	24.73	59.91	53.73	+21.03	79.14
i2b2 2012	Tr	57.09	36.72	65.25	59.08	+1.99	48.05	29.56	71.32	63.40	+15.35	81.10
	Pr	42.93	32.65	53.03	46.87	+3.94	37.49	28.71	52.72	46.57	+9.08	78.97
	Te	51.35	21.58	58.34	47.04	-4.31	29.96	18.15	57.95	49.69	+19.73	72.88
	CD	35.87	19.94	54.85	47.56	+11.69	55.11	11.73	57.17	50.77	-4.34	59.19
CLEF 2014	DD	69.14	34.05	79.01	55.95	-13.19	29.10	16.29	40.85	28.18	-0.92	78.71
i2b2 2018	AD	12.43	4.54	20.93	15.09	+2.66	1.67	2.43	6.20	8.79	+7.12	12.32
	ADE	6.04	1.36	12.36	5.23	-0.81	0.33	0.56	1.76	2.34	+2.01	
Avg.		42.49	23.82	52.02	42.79	+0.30	31.42	18.50	47.75	41.77	+10.35	64.76

Table 9: Extension of Table 1 for Recall (%).

Dataset	Entity Type	UniNER					GNER					UniNER-all (Supervised)
		B	ED	F	EDF	Δ	B	ED	F	EDF	Δ	
i2b2 2010	Tr	56.18	77.13	48.57	64.98	+8.80	63.25	71.70	54.33	60.18	-3.07	70.02
	Pr	55.56	65.28	49.87	56.64	+1.08	50.39	60.12	46.83	54.03	+3.64	70.55
	Te	44.81	63.84	31.02	40.32	-4.49	43.32	68.94	27.01	39.88	-3.44	66.76
i2b2 2012	Tr	52.11	70.34	44.37	57.45	+5.34	52.95	62.61	45.61	51.92	-1.03	65.34
	Pr	50.99	60.64	46.65	54.12	+3.13	45.50	56.20	42.64	51.29	+5.79	71.70
	Te	41.30	58.42	35.29	45.65	+4.35	37.15	57.18	31.83	43.98	+6.83	59.43
	CD	49.04	88.78	25.18	32.56	-16.48	63.20	79.88	29.83	30.03	-33.17	35.49
CLEF 2014	DD	35.29	70.62	31.71	60.75	+25.46	13.10	29.99	11.90	25.74	+12.64	52.79
i2b2 2018	AD	40.93	77.74	39.50	71.63	+30.70	30.34	34.65	26.93	30.70	+0.36	17.24
	ADE	22.86	52.88	22.27	48.11	+25.25	3.78	16.90	3.78	15.31	+11.53	34.00
Avg.		44.91	68.57	37.44	53.22	+8.31	40.30	53.82	32.07	40.31	+0.01	54.33

Test: medical test, laboratory procedure, diagnostic procedure

B Datasets

We include all entities for i2b2 2010, ClinicalIE, and CLEF 2014. For i2b2 2012, we found that UniversalNER and GNER performed poorly on the last two entities (e.g., evidence and occurrence) and decided to exclude them. We attribute this to them consisting mostly of verb phrases, while the training dataset consists mainly of noun entities. For i2b2 2018 Task 2, we test our method on a more challenging setup, extracting *adverse drugs* and *adverse drug events* (Henry et al., 2020). We provide the dataset statistics in Table 7.

C Recall and Precision Performance

We provide the precisions and recalls for each dataset and entity type from Table 1 in Table 8 and Table 9 respectively. We observe a similar trend for both metrics. Furthermore, we observe that UniNER benefits more from precision and GNER on recalls using our framework.

D Filter Prompt

We experiment with different ways to prompt in Section 5.3.4 and provide the specific instructions here.

D.1 Without Description (Default)

Can '**{entity}**' be considered a/an **{entity_type}**? Answer with yes or no.

D.2 With Description

Treatment: Can '**{entity}**' be considered a procedure or substance given to a patient to resolve a medical problem? Answer with yes or no.

Problem: Can '**{entity}**' be considered an observation thought to be abnormal or caused by a disease? Answer with yes or no.

Test: Can '**{entity}**' be considered a procedure or measure to find more information about a medical problem? Answer with yes or no.

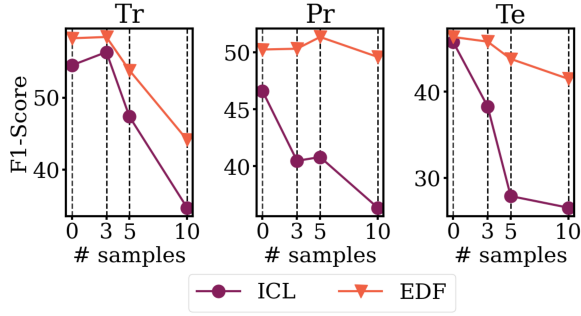


Figure 6: **Few-shot performance comparison.** We observe performance drop using in-context learning (ICL). In contrast, our method (EDF) is more robust. We use the i2b2 2012 dataset with entity types treatment (Tr), problem (Pr), and test (Te).

Clinical Department: Can '{entity}' be considered a clinical unit or clinical service name? Answer with yes or no.

E Few-shot Experiment

Here, we tried including some annotated samples in our framework and compared the approach to standard in-context learning. We randomly sample from the annotation guidelines and add them to the UniversalNER prompt. We also guarantee that there is at least one sample without entities of interest (e.g., sentence does not contain treatments or medical problems). Interestingly, we observe performance degradation across entity types the more samples we use. These contrastive results to general LLMs (Xie et al., 2023) further justify that open NER LLMs cannot be treated similarly to them. Furthermore, we observe that this also applies to our framework, although it is not as severe as standard in-context learning.

We remark that performance drops on in-context learning are not uncommon. Previous works (Zhao et al., 2021; Zhu et al., 2023) show instability in performance for in-context learning. In addition, few-shot experiments are uncommon for zero-shot NER task (Zhou et al., 2023; Ding et al., 2024; Zaratiana et al., 2024), even if they use LLMs. Our work reveals that open NER LLMs may not benefit from in-context learning and are different from general LLMs. We leave further investigation to future works.

F Performance on BERT-based Models

We use GLiNER (Zaratiana et al., 2024), a BERT-based model for open-named entity recognition. Note that previous prompt engineering methods cannot be applied here. We conduct the experiment similar to UniNER and GNER, with Xie *et.al.* (Xie

Table 10: Performance on GLiNER.

Dataset	Entity Type	Metric	GLiNER				
			B	ED	F	EDF	Δ
i2b2 2010	Tr	P	52.03	35.70	70.79	66.71	+14.68
		R	44.55	76.13	39.86	63.93	+19.38
		F1	48.00	48.61	51.00	65.29	+17.29
	Pr	P	71.19	48.48	79.13	67.32	-3.87
		R	49.22	63.36	46.16	56.49	+7.27
		F1	58.20	54.93	58.31	61.43	+3.23
	Te	P	42.80	22.93	63.77	56.65	+13.85
		R	27.23	55.63	23.43	39.14	+11.91
		F1	33.28	32.47	34.27	46.30	+13.02
i2b2 2012	Tr	P	53.77	38.88	71.83	66.92	+13.15
		R	48.37	69.76	42.87	58.03	+9.66
		F1	50.93	49.93	53.69	62.16	+11.23
	Pr	P	71.67	51.93	77.68	67.35	-4.32
		R	50.33	63.93	47.27	58.06	+7.73
		F1	59.13	57.32	58.78	62.36	+3.23
	Te	P	43.97	19.72	66.72	55.42	+11.45
		R	39.17	60.09	35.17	48.41	+9.24
		F1	41.13	29.69	46.06	51.68	+10.25
CLEF 2014	DD	P	48.69	22.99	58.28	50.08	+1.39
		R	71.59	88.27	29.52	32.96	-38.63
		F1	57.96	36.48	39.19	39.76	-18.20
	AD	P	65.32	41.83	72.26	59.42	-5.90
		R	27.90	48.17	26.00	42.99	+15.99
		F1	39.09	44.78	38.24	49.89	+10.80
	ADE	P	2.31	3.52	6.47	13.50	+11.19
		R	5.39	67.15	5.39	61.40	+56.01
		F1	3.23	6.69	5.88	22.13	+18.90
i2b2 2018	ADE	P	7.42	2.17	15.03	7.70	+0.28
		R	14.31	44.93	13.12	40.95	+26.64
		F1	9.77	4.15	14.01	12.96	+3.19
	Avg.	P	45.92	28.82	58.20	51.11	+5.19
		R	37.81	63.74	30.88	50.24	+12.43
		F1	40.10	36.51	39.94	47.40	+7.29

et al., 2023) as our baseline. We present the results in Table 10. We observe the same trend as in Section 5.2 with an average of 7.29% F1-score improvement.

G Performance Drop on CD

We observe significant recall drops to “clinical department” entities across all models. Here, we posit that some entities may not necessarily conform to the clinical department in the clinical domain. For instance, some entities are hospitals; thus, if a filter is prompted with our template (e.g., “Can hospitals be considered as a clinical department?”), then it is likely to reject them. One possible solution

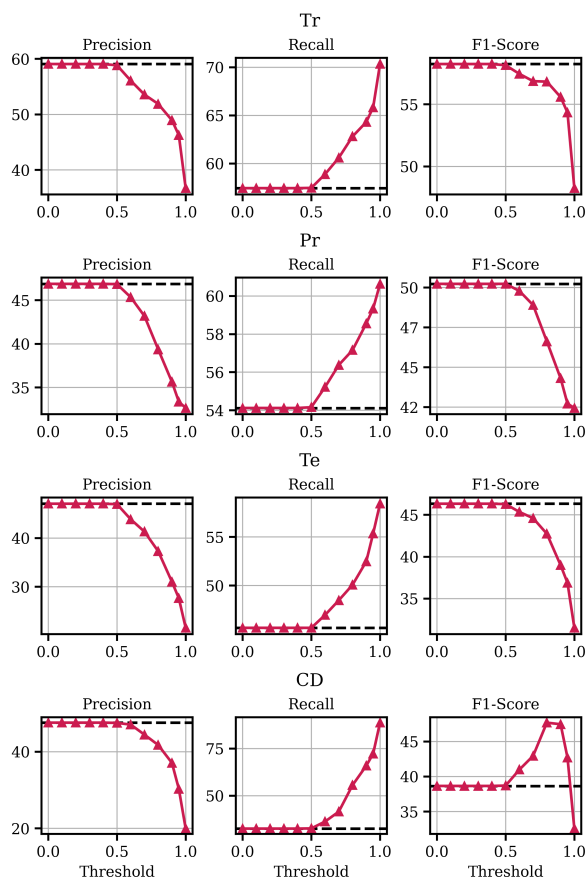


Figure 7: Filter Precision/Recall Trade-off. There is an improvement in recall but a decrease in precision when increasing the threshold. The dashed line corresponds to performance with threshold = 0. We use i2b2 2012 dataset.

is using a clear entity description. As illustrated in Table 5, our framework outperforms the baseline (e.g. 45.97% vs 38.66% F1-score respectively) when using entity description.

H EDF Filter Precision/Recall Trade-off

We analyze how filtering can be made more or less strict to achieve better trade-offs. We use the filter output probability to determine whether the entity is rejected or not. Concretely, rather than directly rejecting them if the filter outputs “No”, we first look at the token probability. If it is less than a certain threshold, we then reject them. Our framework is simplified to entity decomposition if the threshold is 1. We provide the results in Figure 7.

Overall, increasing the threshold leads to decreased precision and improved recall. Interestingly, better thresholding can improve the F1 Score in “clinical department” entities. This might be due to the noises for the entities as described in Appendix G.

I LLM Prompt Templates

Our experiments involve large language models, which are often trained with specific templates. We use their default templates (except Llama2) throughout the experiments and present them here.

I.1 UniNER

A virtual assistant answers questions from a user based on the provided text.

USER: Text: **{input}**

ASSISTANT: I’ve read this text.

USER: **{instruction}**

ASSISTANT:

I.2 GNER

[INST] Please analyze the sentence provided, identifying the type of entity for each word on a token-by-token basis. Output format is: word_1(label_1),

word_2(label_2), ...

We’ll use the BIO-format to label the entities, where:

1. B- (Begin) indicates the start of a named entity.
2. I- (Inside) is used for words within a named entity but are not the first word.
3. O (Outside) denotes words that are not part of a named entity.

{instruction}

Sentence: **{input}** [/INST]

I.3 Asclepius

You are an intelligent clinical language model.

Below is a snippet of patient’s discharge summary and a following instruction from healthcare professional.

Write a response that appropriately completes the instruction.

The response should provide the accurate answer to the instruction, while being concise.

[Discharge Summary Begin]

{input}

[Discharge Summary End]

[Instruction Begin]

{instruction}

[Instruction End]

Table 11: Results from The Original Competitions

Dataset	Metric	SOTA [†]
i2b2 2010	Exact F1	85.20
i2b2 2012	Span F Measure	92.00
i2b2 2012	Type Accuracy	86.00
i2b2 2018 Task 2	Lenient F1	58.73
CLEF 2014	-	-

[†] values reported from the best performing method in the challenges

I.4 Llama2

```
<s>[INST] <<SYS>>
```

You are an intelligent clinical language model.

Below is an instruction from healthcare professional.

Write a response that appropriately completes the instruction.

The response should provide the accurate answer to the instruction, while being concise.

```
<</SYS>>
```

```
{instruction} [/INST]
```

J LLM Hyperparameters

We use the default hyperparameters for each model. For UniNER and GNER, we use greedy search. For Asclepius and Llama2, we use temperature 0.2 and top P probability 0.95. Our exploration reveals consistent outputs for this set of hyperparameters.

K Results from Original Challenges

Each of the datasets in our experiments is curated from a competition (Uzuner et al., 2011; Sun et al., 2013; Henry et al., 2020; Mowery et al., 2014). We present the results from the original competitions in Table 11. The state-of-the-art methods from these competitions are trained using supervised learning. We remark that each competition is evaluated differently, and only i2b2 2010 use the same metric as ours.